

M. Thufail Alwannabil Samas

Data and AI Professional | Machine Learning, Data Systems and Applied AI

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SUMMARY

Data and AI professional experienced in building end-to-end data pipelines, classification and forecasting systems, and applied AI automation. Delivered Python/PostgreSQL monitoring pipelines, XGBoost and BiLSTM models, FastAPI services, Docker-based inference, and local LLM workflows with human-in-the-loop controls. BNSP-certified Data Scientist and first author of a peer-reviewed XGBoost rainfall forecasting publication.

EXPERIENCE

PT Aplikasi Lintasarta

Surabaya, Indonesia

Junior Programmer

January 2026 - April 2026

- Developed an end-to-end DWDM monitoring and assurance system for approximately 400 configured directional sensors, integrating 5-minute PRTG telemetry with ENIMS network metadata to detect active failures, persistent optical-signal changes, gradual degradation, and prioritized conditions for operations review.
- Built an incremental Python and PostgreSQL pipeline for metadata synchronization, telemetry collection, gap repair, quality-controlled observations, event detection, opposite-direction sensor correlation, checkpoint preservation, current sensor-priority updates, and recoverable processing across repeated collection cycles.
- Implemented sensor-adaptive analytics with robust 6-hour medians, PELT change-point detection, Theil-Sen slope estimation, and serial-correlation-corrected Mann-Kendall testing to distinguish persistent downward shifts and gradual degradation from failures, missing data, and short-term telemetry noise across sensors.
- Created a Python API and browser interface with Operations Board, Event Explorer, and Management Overview views, enabling operations teams to inspect active failures, historical events, paired-sensor evidence, investigation charts, and prioritized sensor conditions within one interface for consistent operational review.

SMP GIKI 3 Surabaya

Surabaya, Indonesia

Coding and AI Teacher

August 2025 - December 2025

- Taught Python fundamentals to Grade 8 and 9 students through structured lessons and hands-on exercises covering variables, data types, operators, conditionals, loops, and practical programming logic.
- Introduced artificial intelligence concepts with Teachable Machine, guiding students to prepare datasets, train models, and run image-classification experiments showing how examples become predictions.
- Developed lesson materials, coding exercises, quizzes, sample programs, and structured assessments that reinforced classroom instruction, guided hands-on practice, and evaluated student understanding.

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA)

Surabaya, Indonesia

Programming Teaching Assistant

August 2022 - July 2024

- Guided undergraduate students across 4 semesters of programming courses in MATLAB and Python, using practical exercises and assignment support to strengthen computational problem-solving.
- Supported students in debugging, data handling, and algorithmic thinking, helping them translate mathematical and computational concepts into working code for course exercises and assignments.
- Developed introductory machine learning materials covering Linear Regression, K-Nearest Neighbors, and K-Means clustering, and provided structured technical feedback on student assignments.

Stasiun Meteorologi Maritim Kelas II Tanjung Perak Surabaya (BMKG)

Surabaya, Indonesia

Data Science Intern

February 2024 - June 2024

- Developed a 1-page extreme-weather early-warning dashboard using HTML, CSS, JavaScript, PHP, MySQL, and periodic API updates to centralize forecast data for the BMKG operational workflow.
- Processed and compared BMKG cloud-prediction imagery with Himawari IR satellite imagery using Python to assess cloud movement, temperature patterns, and differences across both imagery sources.
- Built a 3-layer BiLSTM rainfall forecasting model with 7-day multivariate sequences, 27 Grid Search configurations, and an 80:20 chronological split, achieving MAAPE 0.8073 and RMSE 10.2734 mm.

EDUCATION

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA)
Bachelor of Mathematics (S.Mat.)
• **GPA:** 3.46/4.00
• **Undergraduate Thesis:** Rainfall Forecasting in Sumenep Regency Using XGBoost and Grid Search.
• **Academic Highlight:** Completed the 4-year bachelor's program in 7 semesters (3.5 years).
• **Relevant Coursework:** Artificial Intelligence, Programming, Mathematical Statistics, Statistical Methods, Multivariate Analysis, Numerical Methods, and Applied Linear Algebra.

Surabaya, Indonesia
August 2021 - January 2025

SKILLS

- **Programming & Query Languages:** Python, SQL, R, MATLAB
- **Data Engineering & Databases:** Pandas, NumPy, PostgreSQL, MySQL, pgvector, psycopg, Data Ingestion, Data Validation, Data Transformation, Incremental Data Pipelines
- **Machine Learning & Statistics:** scikit-learn, XGBoost, TensorFlow/Keras, statsmodels, ARIMA, Feature Engineering, Statistical Analysis, Model Evaluation, Cross-Validation, Time-Series Analysis & Forecasting
- **AI Engineering & Automation:** FastAPI, n8n, Ollama, RAG, Embeddings, Vector Search, Structured LLM Outputs, REST APIs, Human-in-the-Loop Workflows, Idempotency, Retry and Recovery Workflows
- **Development, Delivery & Visualization:** Docker, Docker Compose, Git, GitHub, GitHub Actions, CI/CD, Unit Testing (unittest), Jupyter Notebook, Model Serving, Batch Inference, Matplotlib, HTML, CSS, JavaScript

PROJECTS

AI Service Request Automation
Short bilingual IT requests required manual triage, so a local AI automation system was built with FastAPI, n8n, PostgreSQL/pgvector, Ollama, and Docker for classification, catalog matching, grounded retrieval, deterministic validation, human approval, retry, and recovery. In the frozen 40-case acceptance gate, the system achieved 100% classification macro F1, required-field accuracy, retrieval Recall@3, and citation validity; 96.7% route and semantic task success; and 90% of semantic requests progressed without service-agent triage review.

Retail Sales Forecasting & Planning System
Retail planning required consistent 16-day forecasts across 54 stores and 33 product families, so an end-to-end pipeline was developed on 3,000,888 labeled rows using forecast-safe features to evaluate 3 Ridge and 27 XGBoost configurations over 4 chronological folds. The selected V2 XGBoost model achieved 0.52 mean RMSLE, 15.24% mean WAPE, and 14.67% protected internal-test WAPE, generated 28,512 forecasts, passed 78/78 checks, and maintained exact prediction parity across batch, FastAPI, and Docker execution.

Thyroid Cancer Recurrence Classification
Thyroid cancer recurrence classification required a reproducible workflow for mixed clinical data and controlled feature selection. The resulting XGBoost pipeline used 364 retained records after removing 19 exact full-row repetitions from the 383-row UCI dataset, applied type-specific preprocessing, fold-local Information Gain, exhaustive Grid Search, and shuffled 10-fold K-Fold cross-validation, and achieved 97.25% pooled out-of-fold accuracy, 95.37% F1-score, 98.05% specificity, and 98.60% ROC-AUC in the completed V1 evaluation.

PUBLICATIONS

Prediksi Curah Hujan di Kabupaten Sumenep Menggunakan Metode Extreme Gradient Boosting (XGBoost) dan Algoritma Grid Search
First Author | MIND Journal, Vol. 11, No. 1, pp. 30-43 | June 2026 | DOI: 10.26760/mindjournal.v11i1.30-43

CERTIFICATIONS

Data Scientist - Badan Nasional Sertifikasi Profesi (BNSP) | September 2024 - September 2027